

# MicroSense AI-Cam Report

AI-assisted optical monitoring report for preliminary screening of microplastic-like particle candidates in water samples.

## 1. Sample Summary

|                       |                       |
|-----------------------|-----------------------|
| Report Generated      | 08 Jun 2026, 10:02 AM |
| Sample ID             | 13                    |
| Sample Source         | Tap Water             |
| Chamber Volume        | 50 ml                 |
| File Type             | image                 |
| Monitoring Risk Level | Low                   |

## 2. Detection and Monitoring Results

|                             |                                  |
|-----------------------------|----------------------------------|
| Detected Candidates         | 4                                |
| Estimated Particles / Litre | 80                               |
| MSMI Score                  | 16                               |
| MPI Score                   | 16                               |
| Confidence Score            | 75                               |
| Particle Size Category      | 100–500 µm estimated (prototype) |
| Average Particle Area       | 1273.50                          |
| Average Brightness          | 43.93                            |

## 3. Hybrid Validation Results

|                          |                                                                         |
|--------------------------|-------------------------------------------------------------------------|
| Raw Detection Count      | 4                                                                       |
| Accepted Candidate Count | 4                                                                       |
| Rejected Candidate Count | 0                                                                       |
| Hybrid Validation Score  | 76.22                                                                   |
| Filter Summary           | All 4 detected particle candidates passed the hybrid validation filter. |

## 4. Image Quality Assessment

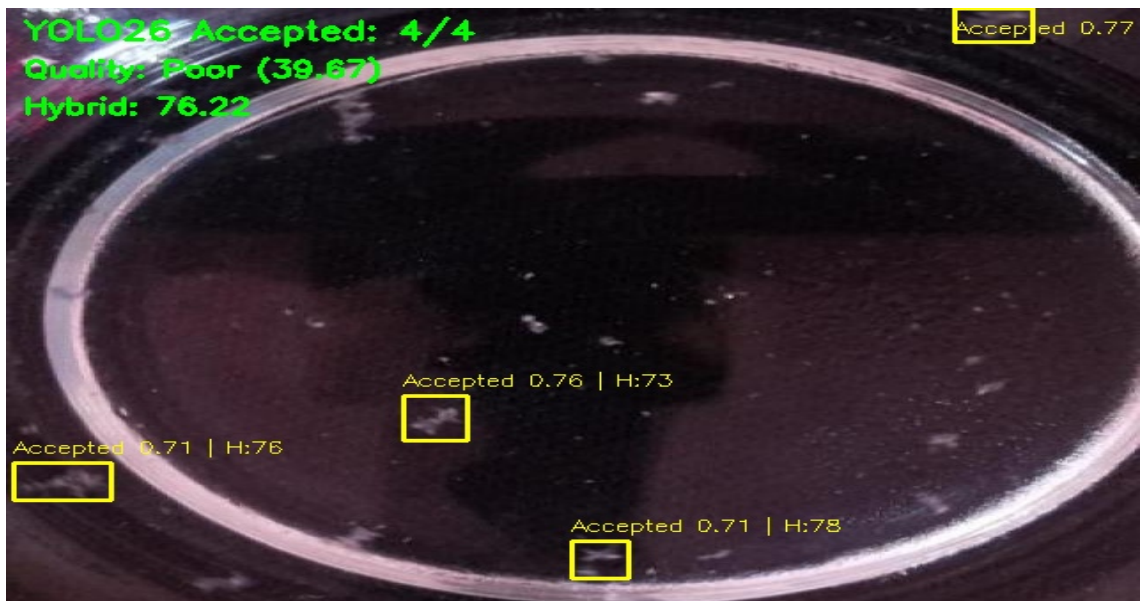
|                      |                                                                      |
|----------------------|----------------------------------------------------------------------|
| Image Quality Status | Poor                                                                 |
| Image Quality Score  | 39.67                                                                |
| Focus Score          | 32.29                                                                |
| Brightness Score     | 32.97                                                                |
| Contrast Score       | 54.33                                                                |
| Overexposed Percent  | 0.44                                                                 |
| Underexposed Percent | 41.48                                                                |
| Quality Warning      | Image may be blurry. Image is too dark. High underexposure detected. |

## 5. Risk Interpretation

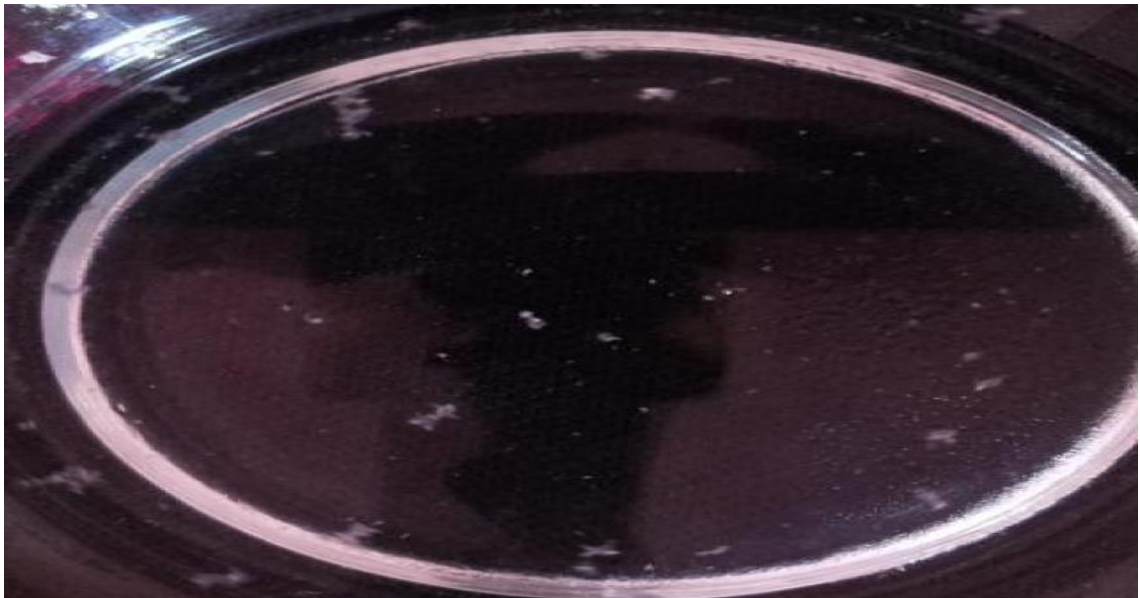
|                               |                                                                                                                              |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Source Risk Factor            | 1                                                                                                                            |
| Concentration Score           | 50                                                                                                                           |
| Size Score                    | 40                                                                                                                           |
| Concentration-Only Risk Level | Moderate                                                                                                                     |
| Risk Explanation              | MSMI is calculated using estimated particle concentration, detected count, average particle size, and particle shape factor. |
| Recommendation                | Low monitoring concern. Continue normal monitoring and retest periodically.                                                  |
| Notes                         | Analyzed using YOLO26n main detector                                                                                         |

## 6. Image Evidence

Processed Output Image



Original Uploaded Image



## 7. Scientific Scope and Disclaimer

This report is generated by MicroSense AI-Cam, a prototype-level AI-assisted optical monitoring system. The reported objects represent microplastic-like particle candidates identified from image-based visual features. The system does not chemically confirm polymer composition and does not replace certified laboratory techniques such as FTIR spectroscopy, Raman spectroscopy, or advanced microscopic analysis. The generated values, including estimated concentration, MSMI score, confidence score, and monitoring risk level, should be interpreted only as preliminary screening indicators and not as regulatory-grade or laboratory-certified measurements.

## 8. System Pipeline

Water sample image acquisition → YOLO26n-based candidate detection → OpenCV-based fallback support where required → hybrid visual validation → image quality assessment → monitoring score calculation → database storage → dashboard and report generation.